MERIDIAN

A live Risk Intelligence Engine: from news to structured risk signals, then to tactical rebalancing and portfolio stress testing

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Problem and approach

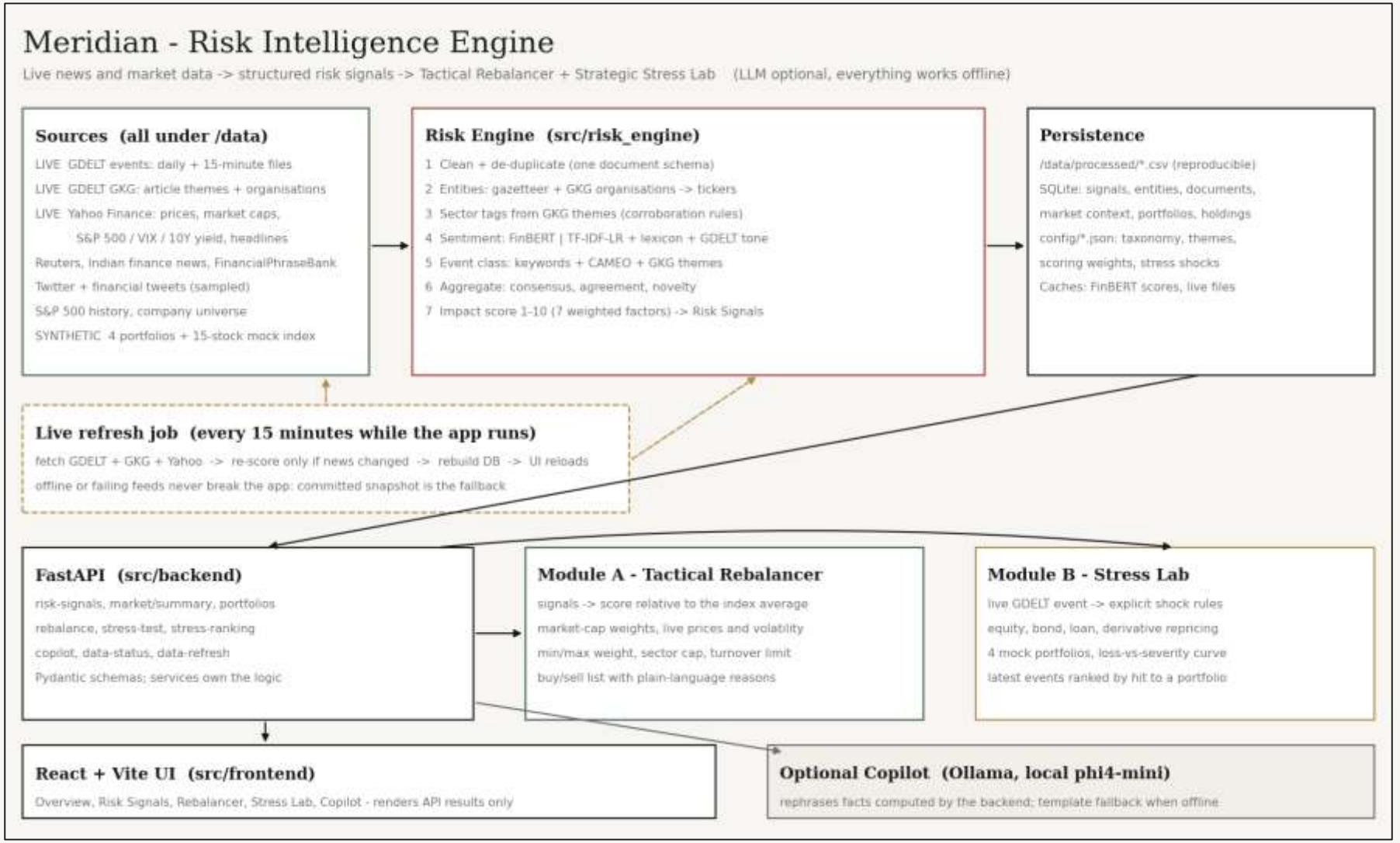
The problem. Risk signals are buried in unstructured text. Portfolio teams need structured, explainable, portfolio-specific inputs, fast.

Our answer. An AI/NLP Risk Engine plus BOTH downstream modules: a Tactical Rebalancer and a Strategic Stress Lab.

The data. Live GDELT events (with GDELT's own article themes and organisations), Reuters, Indian finance news, FinancialPhraseBank, tweets and Yahoo Finance headlines, normalised into one Risk Signal schema.

The rule. Every number is calculated by backend code from data and configuration. A local LLM (optional) only rephrases computed facts and never calculates.

System design



Implementation highlights

Live data. GDELT daily and 15-minute files, GKG themes/organisations and Yahoo Finance refresh every 15 minutes while the app runs, with an offline snapshot fallback.

NLP. Gazetteer plus GKG organisation linking; FinBERT + local TF-IDF-LR + risk lexicon + GDELT tone for sentiment; keyword + CAMEO + theme event classifier.

Impact score 1-10. Seven explainable factors: severity, sentiment strength, confidence, relevance, source consensus, volatility regime, novelty. Polarity alone never sets it.

Rebalancer. Signals scored relative to the index average, then limits on weight, sector and turnover. Live market-cap weights and prices.

Stress Lab. Explicit shock rules in config; stocks, bonds, loans, options, swaps and CDS repriced; loss-versus-severity curve; latest events ranked by hit to a portfolio.

Engineering. FastAPI + Pydantic + SQLite; React only renders API results; automated tests; optional local copilot (Ollama phi4-mini).

Key results

Scale. 50,154 documents became 5,335 structured risk signals (27 Sep - 10 Oct 2026); 20 with impact 7 or higher.

Sentiment. Held-out accuracy: local TF-IDF-LR 0.74 (macro-F1 0.73); zero-shot FinBERT 0.56 (0.54).

Overview. Market risk index, sentiment, top sector from GDELT's own tags, a 10-day comparison and live S&P 500 / VIX / 10Y yield.

Rebalancer. A plain buy/sell list with reasons, real market-cap weights and live prices; turnover stays within its limit.





Domain impact

Early, explainable, portfolio-specific warnings: from a live headline to the expected loss in one click.

The same event hits our four mock portfolios very differently (roughly 8% for the growth fund, under 1% for the conservative one): risk depends on exposure.

Teams can audit every score and shock: factor contributions, source consensus, and assumptions kept in config files.

The break-even view answers: how severe must an event be to cost this portfolio 5%?

Runs fully offline and without an LLM, so it suits controlled environments.



Features at a glance

01

Live news into risk signals

GDELT events every 15 minutes, enriched with article themes and organisations. Each signal gets sentiment, an event type and an impact score from 1 to 10, with the reasons shown.

02

An overview desk

Market risk index, news sentiment and the sector under most pressure, a 10-day comparison you can click through, and live S&P 500, VIX and 10-year yield.

03

Tactical Rebalancer

A plain buy and sell list for a 15-stock index, each with a reason, using real market-cap weights, weight and sector limits, and a turnover cap. Weights shown day by day.

04

Strategic Stress Lab

Pick a live event and see what it would cost four mock portfolios: loss by asset type, sector and driver, a severity curve with break-even points, and the latest events ranked.

05

AI risk copilot

Ask in plain English in a chat. A local model words the answer, the numbers come from the backend, and the data behind each answer is shown as charts.

06

Always current, easy to run

Refreshes itself every 15 minutes, keeps working offline from a saved snapshot, needs no API keys, and every assumption sits in a config file.

Demo video: https://vimeo.com/1234679155 | Setup, data sources and API: see the README